

# Denise Holt: "The Potential Impact of Decentralized AI through Active Infer

55:53

okay hi welcome back welcome back hi

Denise hi thank you for for joining and

looking forward to your presentation so

take it away oh thank you so much and

thank you for having me today um today I

am going to talk about the potential

impact of decentralized AI through

active inference and spatial web

Technologies and what we're really going

to be discussing is um the potential

aspects of this next era of computing uh

it'll be incorporating a lot of the

research that you're hearing about here

over the next three days um and

decentral in my talk decentralized AI is

kind of taking a main focal point here

and there's a couple of aspects um in

technologies that are evolving right now

that uh I think it's really important to

be aware of um and they factor into uh

what I'm going to be talking about today

so one of the main

uh things that is occurring right now is

our internet is evolving um we have a

new protocol layer that's layering right

on top of the other protocols that are

part of the same internet that we're

talking across today um the difference

is is what these protocols hstp

hyperspace transaction protocol and

hyperspace modeling language is the

programming language um what these are

going to enable is for us to have

programmable spaces so instead of just

programming pages in domains with

content we are going to be able to program digital twin spaces of anything and this is going to enable context to be baked into everything in every space so all of the attributes about various things in various spaces and nested ecosystems and all of their inter relationships to each other so one of the things this is going to enable is uh active inference agents to be distributed across our internet and I see this as the next wave of innovation and through the through active inference these agents will be aware of each other and the network uh they'll be able to sense the world in real time and learn from each other and this is going to provide unprecedented levels of interoperability um so what we're talking about here is first principles uh automated systems and distributed collective intelligence so first let's look at the challenges in traditional centralized AI structures um or architectures uh centralized processing constraints you have high energy demands and scalability issues limited adaptability to real time changes and you know we're all familiar with the limitations of uh these deep learning models they require massive amounts of labeled data and processing power they have limited ability to generalize across different tasks and environments AI models uh lack adaptability to real world complexity they're only good for one thing at a time uh like one uh

optimized for one type of an output and  
they lack interpretability and  
transparency in decision  
making they have high computational  
resources leading to high costs and  
inefficiency and they struggle with  
generalization across different scales  
and

tasks now we need decentralized AI and  
decentralization really matters because  
it increases resilience and scalability  
for autonomous systems it reduces  
Reliance on Central servers and data  
hubs and it enables autonomy across  
distributed Networks

so this Bridges the Gap in AI we move  
Beyond centralized AI for scalable  
autonomous systems that can operate  
independently and

locally now we're all familiar with this  
guy here Dr Carl Friston and uh he he  
gave a great talk uh first thing this  
morning on uh renormalizing General  
generative models um so what I'm going  
to be talking about are various  
components in in some emerging  
Technologies along with active inference  
that potentially realize this research  
paper designing ecosystems of  
intelligence from first principles now  
you know the this uh paper was published  
in December of

22 and um it really laid out a new  
framework for autonomous intelligence  
that mimics self-organized systems of  
nested intelligence found in nature and  
in in this paper it described a  
cyberphysical ecosystem of natural and

synthetic sense making in which humans are integral participants what we called shared intelligence and this in the paper it says this V this vision is premised on active inference a formulation of adaptive behavior that can be read as a physics of intelligence and which inherits the physics of self-organization and in this context we understand intelligence as the capacity to accumulate evidence for a generative model of one's sensed World also known as self-evident of the abstract for this paper uh they also note we also consider the kinds of communication protocols that must be developed to enable such an ecosystem of intelligences and motivate the development of a shared hyperspatial modeling language and transaction protocol as a first and key steps towards such an ecology so what I'm proposing in this and what is proposed through all of this this uh technology together uh with all of the various parties that are involved in building these components is that the future of AI is shared distributed and multiscale an entirely new kind of autonomous intelligence able to overcome the limits of machine learning AI and as we know uh from understanding active inference this AI is knowable explainable and capable of human governance and it operates in a naturally efficient way there's no big data requirement and it's based on the same mechanics as biological

intelligence it learns in the same way humans so what I want to do is start with the spatial web protocol uh first because this is kind of the um the framework for these agents to be able to have this distributed intelligence structure so what is the spatial web protocol it's just the evolution of our internet into spatial domains um so people refer to it as web 3.0 it's just the next next evolution of the internet but this protocol has become a global standard a new Global standard for the last four years uh this this standard has been in development with the i e uh which is the largest core standards body uh in the world and back in 2020 they deemed the spatial web protocol a public imperative which is their highest designation and this summer it was voted and approved um as the new global standard and it's in the final administrative uh cycle with the i e before it becomes uh public so so there's a couple of things that are happening with this through hsml the programming language for this new protocol all emerging technology whether it's iot or ar VR distributed Ledger technology um you know all of these Technologies they're all going to be able to uh converge across the internet because they will have a common language for interoperability and in that you also have language that AI can understand and uh communicate together and communicate with these various aspects of Technology

as well so this is going to produce the next era of computing and deoe called it back in 2020 so it's the next evolution of the internet and it's going to be empowered by active inference

AI now just to kind of understand what this means for the internet um let's take a look at what a website domain is today with the worldwide web a domain basically contains a website and these websites they serve up content so whether it's you know images or video or audio or you know information data um that is what websites are capable of of doing now a spatial web domain is quite different because a spatial web domain could be anything in any space right and it can be programmable a spatial web domain is an entity with a persistent identity through time that has rights and credentials so you're talking about going from a library of pages in our Internet to the internet of everything and hsmI becomes a common language for everything so it provides unprecedented levels of interoperability between all current emerging and Legacy Technologies uh it's a programming language that Bridges communication between people places and things laying the foundation for a realtime World model for autonomous systems and what you have are programmable spaces hsmI is a cipher for context anything inside of any space is uniquely identifiable and programmable

within a digital twin of Earth producing  
a model for data normalization and from  
this we get adaptive intelligence  
automation security through Geo encoded  
governance multi- Network  
interoperability and it enables all  
Smart Technologies to function together  
within a unified  
system and then this produces a  
Knowledge Graph and digital twins of  
anything everything potentially our  
planet so as as this knowledge graph  
grows within the programming of all of  
these spaces across the network you get  
this Universal domain graph that enables  
a digital twin of our planet and all  
nested systems and entities within it we  
get nested  
ecosystems uh there there will be  
intelligent agents both human and  
synthetic across this network uh they  
will be sensing and perceiving  
continuously evolving environments  
making sense of the changes updating  
their internal model and what they know  
to be true at any given moment and  
acting on the new information that they  
receive so why do we KN need a spatial  
Internet  
Protocol well the increasingly  
graph-like nature of global data the  
opportunity for Automation and uh  
autonomic activities using context aware  
cognitive AI the need for composable  
systems and applications including the  
governance of such systems the intrinsic  
intrinsic need for secure transactions  
the rise of machine learning and neural

network computation and Edge Computing  
the need for explainable AI and robotic  
governance and the rise of IoT and  
sensors  
mesh and some of the guiding principles  
of the spatial web standards are  
spatiality representing information as  
locations and relations in hyperspace  
ownership users own their data in  
digital property security secure data  
collection transmission and storage  
privacy individual control over digital  
identity and personal Data Trust  
reliable permission driven validation of  
users assets and spaces interoperability  
seamless navigation and asset transfer  
across spaces and responsibility  
creating technology ethically for the  
benefit of  
humanity and what has become of  
these Global standards is a socio-  
technical standard so the spatial web is  
a socio-technical system of  
systems that takes into account the human  
so the development of the spatial web  
must address both technological as well  
as sociological considerations socio-  
technical standards integrate and  
balance the technical social physical  
and legal use of technology and the  
spatial web standard is a socio-  
technical  
standard and what this results in is  
multiple scale cognitive Computing so  
the spatial web plays the role of a  
collective nervous system for smart  
infrastructure it hosts cognitive  
Computing human inference perception

learning and action selection by  
combining and abstracting distributed  
information the spatial web enables the  
formation of complex ideas and  
facilitates

decision-making cognitive Computing  
occurs at multiple scales in the spatial  
web individual nodes host cognitive  
Computing functions and networks of  
nodes collectively perform emergent  
cognitive

Computing so hsml hyperspace modeling  
language it's a human and machine read  
readable modeling language and semantic  
data ontology schema there are some key  
elements to it there are entities  
activities agents contracts channels  
credentials domains and it includes  
hyperspace and time so hsml enables the  
expression and automated execution of  
Legal Financial and physical activities  
in the spatial web and all of these  
entities are identifiable and knowable  
by these intelligent agents through this  
uh language throughout the  
network now let's look for a second at  
the agent aspect agents and activities  
so agents and that can be human agent  
agents or synthetic agents um but these  
agents are entities that sense respond  
maintain a model of their environment  
and take actions to achieve goals and  
the activities that occur they're single  
actions or partially ordered sets of  
actions that are performed by the agents  
and they unfold over time so agents  
interact with domains and other agents  
to perform activities in the spatial web

so what the spatial web facilitates is a distributed ecosystem of intelligence if you look at current state-of-the-art AIS they're siloed applications they're built for optimizing specific outcomes and they're they're really unable to communicate their knowledge frictionlessly and collaborate with other

AIS but this facilitates a a an ecosystem of intelligence that is a self-evolving system system it's learning moment to moment and upgrading its World model and in that way it mimics biology while also enabling uh general intelligence so this is shared intelligence at the edge of everything so how active inference works in the spatial web it enables realtime decision making you have agents that are predicting and adapting to their environment from their frame of reference in the network uh they're continuously learning from their environment enabling realtime adaptation and unlike traditional AI which relies on pre-programmed rules and static models active inference continuously updates its models based on new data allowing it to adapt in real time so these active inference agents would seamlessly communicate in real time with each other sharing their knowledge and perspectives and learning from each other these active inference agents excel in handling complex uncertain environments by continuously minimizing uncertainty in fact they can

quantify their level of uncertainty and this makes them suitable for applications where conditions are constantly changing so decentralized AI takes the processing to the edge to the edge devices you have a distribution of process in and power uh superpower gpus are not required the processing can take place on local devices throughout the network with infrastructure that already exists like your laptop or your mobile device or devices that haven't even been um invented yet um but we're already seeing a lot of those kinds of devices coming into play uh with you know um Apple Vision Pro and and you know the metal glasses and and different aspects of these

#### Technologies

so the processing takes place on these local devices throughout the network you have faster response times and reduced Network congestion and it uses real live data real-time data from iot sensors machines and everchanging context that's embedded to the uh in the network so this is the sensory information that's coming in for these active inference agents so that they can uh you know do the perform this action perception feedback loop and continually update their model of the world of their reference and it eliminates the need to tether to a giant database active inference agents throughout the network learn and adapt from their own frame of reference within the network enhance

data privacy by minimizing centralized storage and it minimizes complexity it uses the right data in the moment for the task at hand and with hstp with hyperspace transaction protocol you there's natural guard rails uh included in this framework so it creates a secure ethical and user Centric environment for AI applications fostering trust and reliability in digital interactions uh you have data privacy and sovereignty trans parency and explainability security and authentication bias mtig mitigation ethical AI guidelines interoperability standardization user empowerment and control accountability and governance safety and reliability Dynamic policy enforcement and what this results in is in intelligence at scale uh through this network of distributed intelligence it enables the cognitive architecture of the collective intelligence of multiples of agents that continuously communicate coordinate and collaborate with each other you have individual and specialized intelligences together on a common Network speaking a Common Language

hsml efficient and Powerful cross communication to perform tasks regulate systems and address problems in real time and it scales up and grows in tandem with humans through hsml we can take human laws and guidelines make them programmable for these agents to understand and abide by in real time so

we're talking about unprecedented levels of cooperation between human and machine intelligence and what you get then is an Ever evolving collective intelligence uh decentralization is an imper imperative for intellig at scale these agents represent scores of people from all over the world sharing their individual and specialized knowledge all coming together on a common Network and it's based on each agent's own world model you have unique perspectives from whatever frame of reference uh within the network nested ecosystems and levels of self-organization and Collective shared intelligence so the idea is that anybody can part their knowledge and their wisdom into these Agents from wherever they are in the world and this actually acts to preserve all of the uh cultural differences all of the um the belief differences the governing Styles um you know people whether they're in uh Kenya or Singapore or anywhere else in the world they should be able to create these agents to solve their local problems and then share that with with the world because then these agents can share their intelligence with each other so the spatial web protocol enables a decentralize secure and interoperable web with the potential to profoundly transform Global Industries and everyday human life now let's dive into um kind of the the potential that exists for this idea of rgms these renormalizing generative models and the

spatial web and we heard a great talk this morning uh by Dr friston he uh you know he talked a lot about the you know the science behind these uh these rgms so I'm going to kind of uh maybe maybe break down some of the aspects of how these work as leading up to then what they can do within this uh potential potentially with in this architecture so you know with rgms one of the interesting things is that it's a universal architecture um they're a UniFi unified scale-free framework for AI That's capable of reasoning learning and decision making and they can generalize across multiple tasks handling small and large scale data efficiently they combine perception learning and planning into a single system and they can be used to learn scalable architecture over space and time similar to human thinking they require significantly less training training data than traditional models while achieving Superior results so what are these rgms they're a new class of AI models based on active inference they use renormalization to simplify complex data into hierarchical structures and they operate on the free energy principle to minimize uncertainty and adapt in real time so the significance is that they represent a Leap Forward from traditional deep learning methods they reduce data requirements and uh improve efficiency and it's one single framework for all

applications so the science behind rgms  
renormalization and multiscale learning  
renormalization is a technique from  
physics that's used to simplify complex  
systems to maintain consistency at  
different scales and rgms break down  
data into multiscale  
representations hierarchical structures  
rgms are actively learning from small  
details to large patterns  
simultaneously and they analyze like  
with an example would be like an rgm can  
analyze an image by recognizing edges  
then objects then scenes and  
understanding  
the the concepts at every at every  
scale so key Innovations and advantages  
of rgms why they're Superior to  
traditional models they learn efficient  
efficiently with less data it's a  
unified system that handles perception  
and planning  
simultaneously and they understand  
conceptual relationships not just  
patterns and they can adapt to new  
environments with minimal training uh  
aspects of them are scale-free modeling  
renormalization discrete representation  
hierarchical structures fast structure  
learning unified perception and planning  
and explicit handling of  
uncertainty so if you look at how the  
brain process data processes data um you  
know we we perform core screening all  
the time when we're learning right we  
break down sensory data into hierarchic  
patterns um that help us focus on the  
most relevant

information um kind of this chunking of Concepts to to grasp different aspects of um whatever it is that we're we're learning or focusing on in the moment and rgms can mimic this approach by processing data hierarchically focusing on important patterns so it's a first principles approach uh using a way to simplify models without losing important information um they can model space and time Dimensions like we do and can learn the causal structure of information and it helps handle large scale problems efficiently using less data and less energy now when you consider uh scale-free modeling you know um scale-free modeling these rgms they operate across different scales so from very fine details to highlevel con uh complex systems so unlike deep learning which struggles when moving between small and large scale problems this scale-free architecture allows rgms to seamlessly work on a variety of problems including small data problems and large scale strategic planning so when you when you're thinking of the concept of scale-free you know this is a great example here of a green apple you know you can you can see the green apple a concept of just you know pixels in an image or a volumetric object or on a branch on a tree or in an orchard and you know the the way that these uh rgms can model these conceptual understandings of things is it can

understand all different scales of what a green apple is so when you think of like um the supply chain or or factory operations or or anything like that these agents can understand the what's happening across the manufacturing line on a very uh local level but also what's happening across the entire distribution line uh throughout the entire system so they can predict both short-term movements and long-term outcomes so you know the fine details as well as uh large larger uh outcomes with the highle complex systems all at the same time and they don't lose accuracy when switching between local and global data scales so hierarchical modeling in rgms hierarchical data processing rgms process at multiple levels of abstraction from fine grain details to highlevel

Concepts uh they use recursive learning they continually Loop over data learning at each level continually refining their understanding of Concepts as they process new information and you get hierarchical representations that they build models of the World by zooming in and out of different layers of detail and this is similar to how human cognition processes both micro and macro scales so you know think about when you're when you're thinking about the concept of driving a car if you're talking to somebody about driving a car or you know you're you're focused on the aspect of driving you're not thinking about the car as all of its

various parts that make up that complex system like the brakes and the steering wheel and the seats and the every aspect of the engine you just call it a car right because that's all you need to know in the moment for whatever whatever you're uh doing with that concept and so this is kind of a similar idea um and this layered approach allows rgms to understand complex multiscale problems without requiring vast data sets so conceptual modeling for decision making um rgms then focus on conceptual modeling versus just mere pattern recognition um deep learning can recognize patterns but struggles with conceptual understanding it can recognize correlations without understanding relationships and it can recognize patterns but it doesn't understand the underlying meaning and rgms on the other hand focus on conceptual modeling they capture multiscale hierarchical structured Concepts that represent both fine grained and Abstract features it builds a hierarchical understanding of the world capturing cause and effect relationships so rgms understand the why behind patterns not just the what and the key difference rgms are not just a better generative AI but a transformative AI framework that builds world rgms can reason about cause and effect relationships making decisions based on conceptual understanding that are more aligned with real world

outcomes so how do rgms leverage active inference by incorporating active inference Active Learning and active selection while incorporating realtime data to adapt its predictions and actions so what you get is realtime learning uh they adjust their beliefs based on new sensory input making decisions on the Fly optimizing both learning and decision-making processes and they also are able to uh have unified perception and planning so deep learning typically separates perception like recognizing an object from planning like deciding an action rgms can combine both in a unified model they handle both within the single model and learn Concepts that guide both recognition and future decisions so for example you can have an autonomous robot that can recognize a door handle and also plan how to open it using the same model so how what does all this have to do with the spatial web architecture so the potential that I see with rgms is that um the nature of how they work fits very well with the nature of this architecture within the spatial web and just to kind of bring you back to what I was talking about earlier you have hyperspace transaction protocol this manages data flow between interconnected systems and it facilitates interoperability between different AI systems and the hyperspace modeling language provides a common language for these systems enabling uh computable context and

decisionmaking and it provides a shared framework for context aware decision-making so this protocol aims to create a standardized way for different Technologies and AI systems to interact and share information across programmable spaces within our global internet and this unified model this becomes a unified model for diverse applications handling complex multiscale systems through unified perception and planning it's a single framework that can handle multimodal tasks so you have versatility they can perform across multiple applications whether it's object recognition natural language processing strategic planning uni using a single Universal model and it reduces the complexity uh it integrates the data analysis the ception and the strategy execution which is the planning within one model so you have cost savings it reduces the need for multiple specialized AI models it streamlines system management and saves resources so this how the spatial web enhances the rgm capabilities the combination of this scale-free active inference with the spatial web protocol could enable seamless integration of phys of the physical and digital these protocols allow rgms to operate across distributed networks exchanging insights and information between physical and digital environments enabling more sophisticated

iot uh

applications uh context aware Computing  
through the spatial web rgms would  
operate in context aware environments  
these agents uh adjust their behavior  
based on the phys physical and digital  
surroundings and interoperability  
multiple rgms these active inference  
agents and autonomous systems can  
operate in different environments  
whether it's your Home Smart City  
businesses and they could communicate  
collaborate and coordinate actions  
through the hyperspace transaction  
protocol there uh energy efficient  
distributed intelligence instead of  
relying on centralized energy intensive  
data centers intelligence could be  
distributed across networks of smaller  
more efficient  
devices and this this creates a nested  
ecosystem uh similar to a hallon  
architecture and if you're not familiar  
with a what a hallon is it's something  
that can be both a whole in and of  
itself and also part of a larger system  
so the spatial web and rgm would form a  
nested ecosystem where each node in the  
network whether it's a domain device  
agent or subsystem functions as a hulon  
it's autonomous yet interconnected  
contributing to a larger system's goals  
as part of the larger interconnected  
Network like a smart City  
infrastructure um your smart home could  
contribute the contribute data to your  
neighborhood grid which informs the city  
infrastructure decision all while  
functioning

independently um if you if you look at this picture here of you know uh part of a supply chain image you know each item packed inside of each of those boxes is a hole in and of itself but it's subject to and it's part of the larger system the Box the box is subject to the the trucks that are moving it and distributing it or you know whatever's happening within the factory with those boxes and then that factory itself is its own hole that is subject to a larger ecosystem of the entire supply chain as a whole so when you're looking at what this next era of computing really enables it's just nested ecosystems that can be empowered by this natural intelligence and active inference Ai and the spatial web protocol can work together to create sophisticated adaptive systems at various scales so rgms for autonomy and self-organization with recursivity the scale-free models naturally form a nested hierarchy each level of abstraction in the model is complete in and of itself and yet part of a larger whole uh autonomy and cooperation active inference agents exhibit both autonomy in their decision-making and the ability to cooperate with other agents and self-organization the renormalization technique in introduced in the paper allows the system to dynamically organize information at different levels of conceptualization similar to the self-organizing nature of uh honic

systems or these in these nested  
ecosystems

so this last part um I want to explore  
the transformative potential of  
decentralized AI for the future of  
intelligence

systems so the impact of distributed  
Collective intelligence the impact on  
this next era of computing would be  
enhanced problem solving and Innovation  
increased efficiency and scalability  
democratization of knowledge and  
Technology resilience and robustness  
enhanced decision making personalized  
and context aware Services ethical and  
responsible AI collaborative platforms  
and ecosystems enhanced learning and  
education empowered communities  
accelerated scientific research and  
environmental environmental and  
sustainability

Solutions and the implications for  
future systems and

Society the convergence of rgms in the  
spatial web could transform Industries  
enabling decentralized context aware  
systems that operate with minimal human  
intervention industries that could be  
impacted smart cities you would have  
autonomous systems that improve  
efficiency at all levels Global Supply  
chains rgms could optimize processes  
from local shipping to Global  
networks personal to critical systems  
homeostasis so whether it's personal  
devices or critical infrastructure a  
Continuum of intelligent adaptive  
systems can maintain homeostasis at each

level in our Health Care Systems  
personalized treatment by integrating  
real-time data from wearables and  
medical records education we can have  
Aid driven adaptive learning systems  
that evolve with student needs and  
Environmental Management we can have  
ecological systems that could be modeled  
and managed more effectively from  
Individual habitats to Global Systems  
balancing local interventions with  
global  
goals and the future of AI and human  
civilization the development of  
renormalizing generative models within  
the scope of active inference and the  
spatial web promises to have farreach  
far-reaching  
implications more adaptive AI these  
systems are highly adaptable capable of  
operating in new environments with  
minimal training efficient Edge  
Computing rgms enhance the power of AI  
on devices like smartphones and iot  
sensors reducing the need for large data  
centers improved human AI interaction  
with explicit uncertainty models these  
systems interact naturally with humans  
seeking clarifications as needed safer  
AI systems they recognize and handle  
novel or ambiguous ambiguous situations  
reliably preventing catastrophic  
failures in unfamiliar scenarios so they  
can fail  
gracefully and accelerated scientific  
discovery rgms expedite research in  
field Fields like genomics and climate  
science by rapidly identifying patterns

and formulating plausible hypotheses  
thereby guiding further experimentation  
so the key takeaway here is that uh all  
of the all of this research and all of  
these Technologies together can provide  
us with a future where AI systems can  
interact seamlessly with the physical  
world autonomously optimizing operations  
and improving human lives  
so um yeah there's my website I have a  
podcast and I'll stop sharing  
now thank you Denise awesome I don't  
know what time how how's how are we on  
we're 15 minutes and I'll read some of  
the questions in the live chat okay sure  
I'll start with Pablo who wrote how will  
all of this affect the gamification of  
processes in organization  
s well I mean the the interesting thing  
is is through hsmI you know all of these  
emerging Technologies are are going to  
converge together and become  
interoperable in ways that we've not  
experienced before you know we've had  
this um kind of vision of of this  
augmented reality existence but the  
framework hasn't been there and this is  
going to provide that framework so I  
think when you're talking about about  
gamification in every aspect it'll  
probably become kind of a natural part  
of of our existence and how we move  
through our day how we interact with  
with our work with uh brands with you  
know anything um you know I think that  
there's going to be a lot of of really  
interesting ways to kind of perpetuate  
that cool there's a lot of great

questions in the chat so okay I'll read one from Tintin so they wrote it's impressive the obvious question I have is of alignment of this system in terms of that alien message from elaser gudowski for example and then they clarified what that exactly is so the key idea there is that simulated agents evolve and came to think faster than their creators so at a moment they took control to ensure their own survivability it's the question of how to ensure that way more intelligent and faster systems keep aligned with our values right and so one of the things that uh is really interesting to me about this is through hsmi we can actually collaborate with these systems in the sense that we can take human laws and guidelines and make them programmable so these agents can understand and abide in real time and you know um versus uh who you know Dr Carl friston is is their cheap science scientists um they were involved in a a program called Flying forward 2020 and and they were doing a lot of uh testing around this and because it was it involved a drone project that I I don't know there were multiple countries throughout Europe it was with the European Union that were involved and you know testing whether the they could program in laws around airspace laws and you know things like that they did all kinds of scenarios of like you know Medical Supply delivery uh perimeter security all kinds of things and what

they found was through hsmi you could actually uh program these guidelines and these these agents abide by them so I think we are going to experience this unprecedented level of of cooperation and collaboration so we can grow in tandem with the you know with these uh agents and their intelligence and I think that we can also you know take some um solace in the fact of how you know active inference works and you know how our human knowledge grows you know I think we're going to see a similar pattern with this knowledge among these agents and the only difference is and and this is just me speaking from you know my own you know my own uh perception and and perspective of this but you know they're going they're going to be dealing with real-time data and with real uh a real grounding of this programmed these programmed spaces and those are the things they're going to be operating off of to make their decisions right whereas if you look at a human you know we have so many other variables going on and every when we learn through active inference it's a very subjective uh growth that's why you can have you know multiple children growing up in the same house household and they all experience things differently right but when you have these agents and they're getting this they're getting their learning through actual tangible programmed information I think there's a

a stability there as well if that makes sense and and and and again that's just my own uh you know interpretation and feeling around this and I could be I could be wrong but you know just kind of throwing that out there that's how we move forward

yeah okay next question from David

Highland who

wrote thank you for the informative presentation Denise can you share any thoughts or information about the ownership model for agents is it a decentralized market how do we decide which parts are public

goods yeah so with the protocol with a with hyperspace transaction protocol um because so it it's interesting it really shifts to this uh kind of self-

sovereign identity uh aspect of the internet because right now think about how web domains work you know if you want to interact with a web domain all the transactions are taking place under the umbrella of whatever centralized organization owns that domain and you have two choices you can either say yes you can have all my data and you can decide what to do with it all or I don't want you to but then I'm opting out of being able to engage with your

domain well in the spatial web because now transactions can be at every touch point of every entity within the network then permissions can be programmed in a very nuanced way right so it really puts that back in the hands of the of the user of of the entity within it you can

you can program you can program guardrails around certain aspects of data while allowing permissions and Freer like Freer accessibility for other aspects of data so whether you're talking about a business you know they can guardrail off their data they can guardrail off certain aspects of it for certain Departments of employees some for their customer base some for you know um Partnerships whatever right um and we're going to be able to do that kind of thing too and the thing with this is this takes into account time the passing of time so you can set expirations on permissions as well so I think we're going to have a lot more self- Sovereign control in this in this area yeah that's a part that really excites and Peaks me as well like with the web domain or the email domain it'd be like saying well on one hand we're not going to look in your email let's just go with that for now but we do or don't accept letters from this area so it's kind of all or not nothing and it's based upon like you pointed out the the custodianship of the all or none permission based around what boxes you can send where whereas having a more nuanced way to talk directly about belief sharing means that instead of just like plugging the USB into the port there could be even a active inference negotiation around what elements to share how to share when to expire all

these other features which can help  
like incrementally build rapport and  
add to the relationship instead of doing  
all or none which can be just very prone  
to false positive or false negative in  
cyberphysical

interactions yeah you're so right Daniel  
yeah what do you think about learning  
trajectories and paths I know it's a  
interest of all of ours I guess like  
what what can we do to be learning and  
thinking about all these

things well um so you know I I have I  
actually have put together a series of  
courses um to kind of give people a a  
real

um kind of a prerequisite knowledge  
understanding of what this transition  
looks like and the different components  
and what it means from all aspects of  
like you know basic understanding of  
active inference and uh you know the  
free energy principle and the spatial  
web Technologies to you know

decentralized AI what does that mean  
convergence of all this Tech across the  
network what do that look like safe AI  
how does active inference benefit that  
and then rgms let's let's look a little  
closer at at that and what I try to do  
is I try to take the complex ideas but  
make them more understand  
so that people can really wrap their  
heads around the these Concepts and then  
they can go dive into the more technical  
aspects of it and I know you with the  
Institute you have a lot of technical  
knowledge there so I think that you know

between resources like you know some of my resources your resources and then I'm sure there are resources out there that maybe I'm not even aware of and you know I think there's going to be more and more resources available around this modeling these agents you know uh building out you know these these spaces uh within the network um I know that versus is going to be launching a platform that'll be the first interface for the public around this technology and I think it's coming to the public at the end of first quarter next year and with that there will be tools where people will be able to build these agents as applications with in the network and and you know I think at that point it's going to get really exciting um so yeah I I just think over the next six months we're going to see a lot coming about with this and if anybody is interested in the courses I've put together you know you can you can visit my uh my website and and sign up for them uh they're not available yet but they'll be available very shortly uh within the next week or two so yeah cool okay I'll I'll for last question ask the same thing that I asked uh Alex and Carl earlier what is a fun or a memorable moment from learning and applying active inference this year and then if you haven't already mentioned it what something that you're excited for next year um this year honestly I I think that uh I have been on this Learning Journey in creating this course material

around it you know and at the same time  
I've been building community and kind of  
having them be my test guinea pigs  
around it and stuff so I you know  
there's been this whole learning aspect  
for me around building this educational  
material so it it has been you know a  
constant uh feedback loop and next year  
I'm just very excited about the  
potential of um building these agents  
and what that means and and how we can  
tackle some of the the problems and the  
the issues and the challenges that we  
have um through this new exciting  
technology and this this uh framework so  
yeah well thank you for all of your  
great learnings and digested and and  
sharings and and all the ways that that  
what you talked about today I think  
connected a lot of the dots from hearing  
Carl's more technical view on the rgm on  
through what what does it mean to have  
multi-agent decentralized as opposed to  
just building One agent so thank you  
again Denise thank you Daniel thanks for  
having me thanks everybody till next  
time  
bye e